TIME APPLIANCES (TAP)



Precision Time for AI Infrastructure: Deploying PTP Boundary Clocks at Scale

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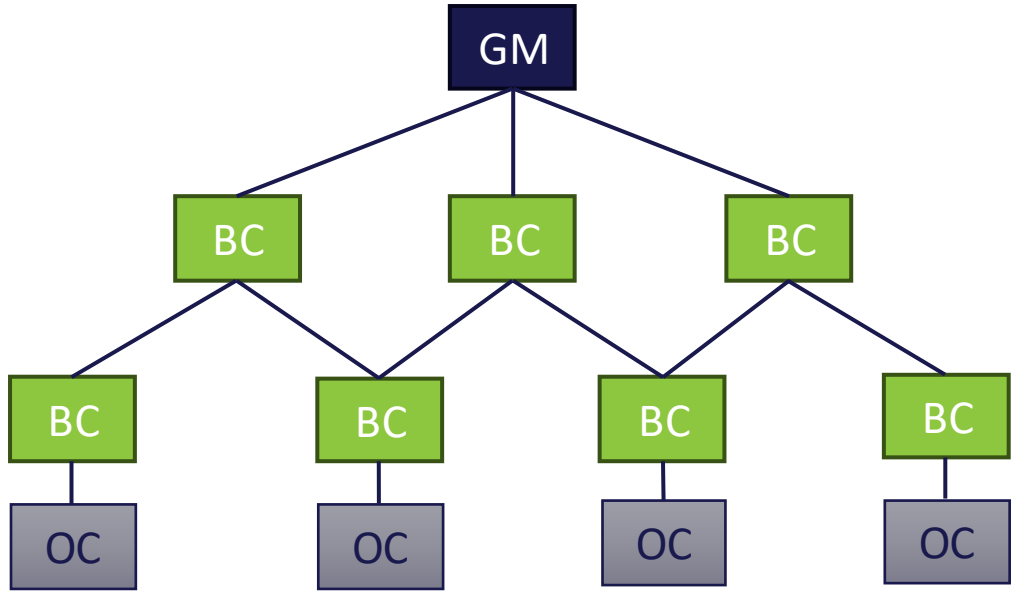
OCP PLATINUM Member

Precision Time Synchronization for AI Ethernet Fabrics

- Distributed AI systems increasingly depend on coordinated timing across:
  - GPU/accelerator nodes
  - Ethernet switches
  - Storage systems
  - Telemetry infrastructure
- Conventional NTP synchronization is not sufficient when network events need sub-microsecond alignment
- IEEE 1588 Precision Time Protocol (PTP) provides the protocol foundation
- Boundary Clock (BC) architectures allow PTP to scale through large Ethernet fabrics
- Hardware timestamping removes much of the non-determinism introduced by software packet processing
- SAI provides a vendor-independent interface for exposing switch capabilities

Boundary Clock in an Ethernet Fabric

- If we only have single grandmaster directly serving every endpoint
  - Large PTP message fanout
  - Greater processing requirements
  - Larger timing domains
- Instead, Each BC
  - Synchronizes its local clock from upstream
  - Becomes a master for downstream
  - Regenerates timing hierarchy

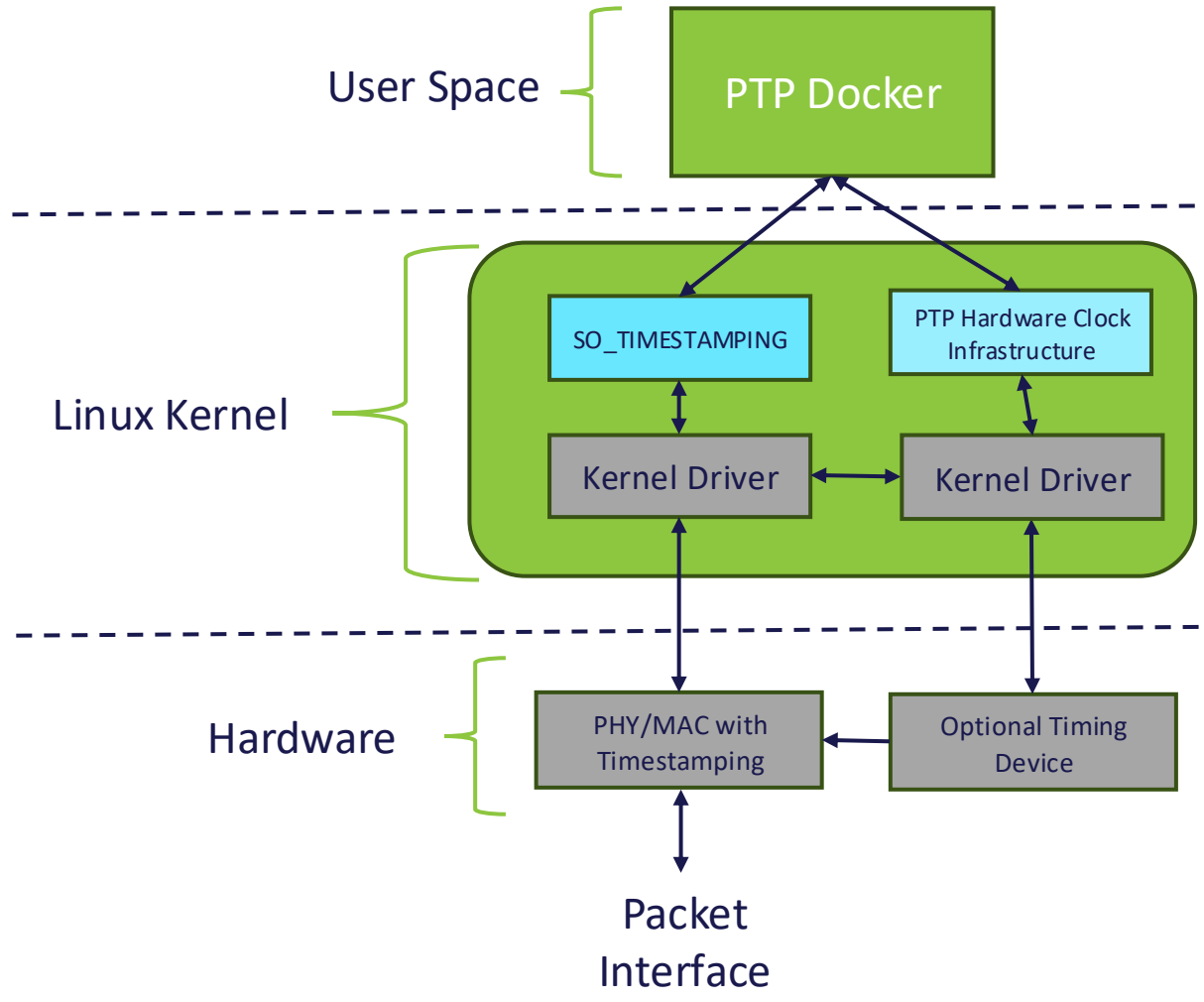


- This brings scaling advantage where the upstream and downstream timing relationship is localized

Sonic Support

- Phase 1 of Sonic specific changes targeted for 202611 release
  - One step hardware timestamping on single ASIC pizza box
  - L3 over UDP
  - Boundary Clock support
- HLD available at https://github.com/sonic-net/SONiC/pull/2324
- Code PRs will be available shortly

Sonic High level Components

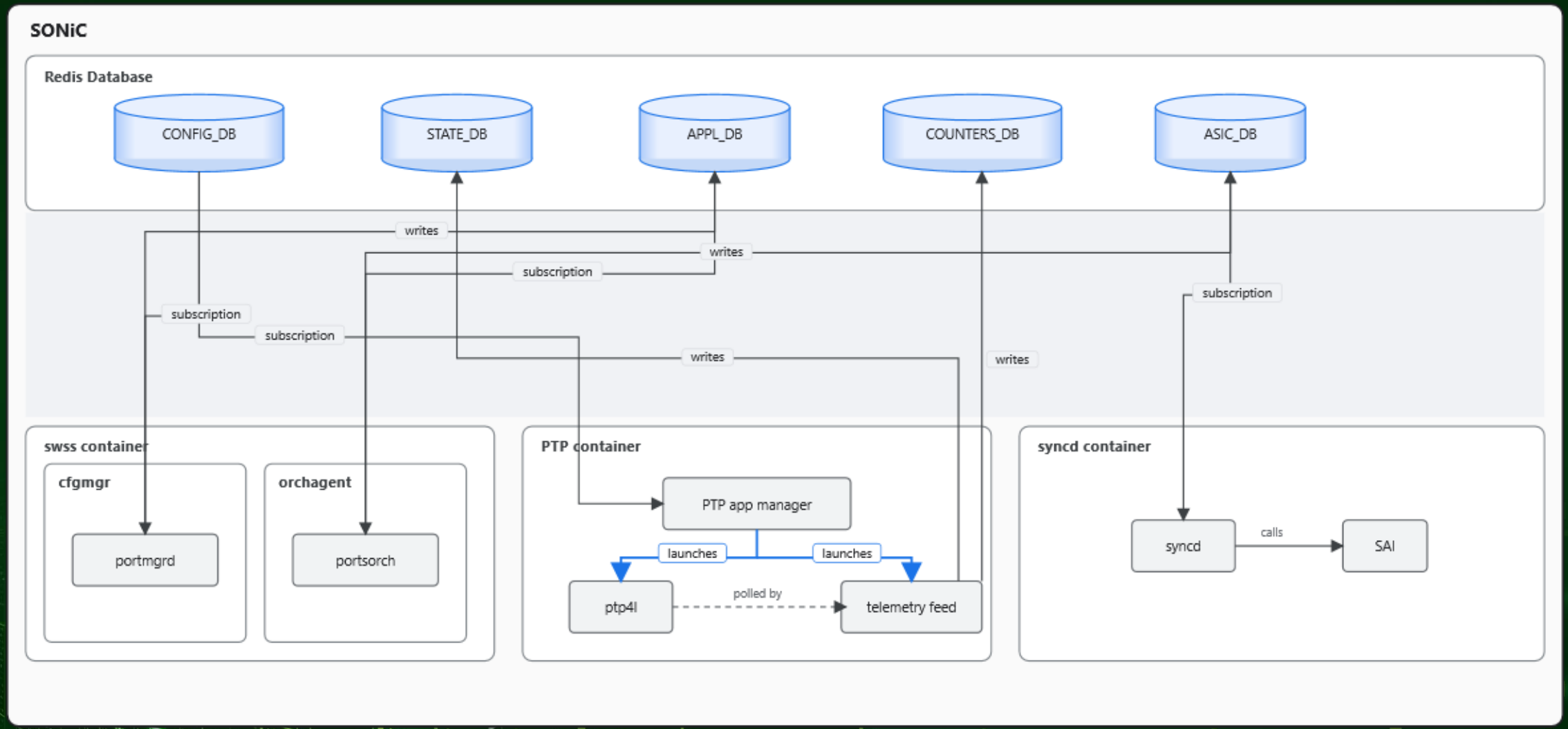


- Includes Linux PTP protocol stack (ptp4l, pmc) exectuables
- Implements BC, OC and TC

- Provides a generic interface between hardware and protocol stack

- Hardware timestamping

Implementation on SONiC Switch



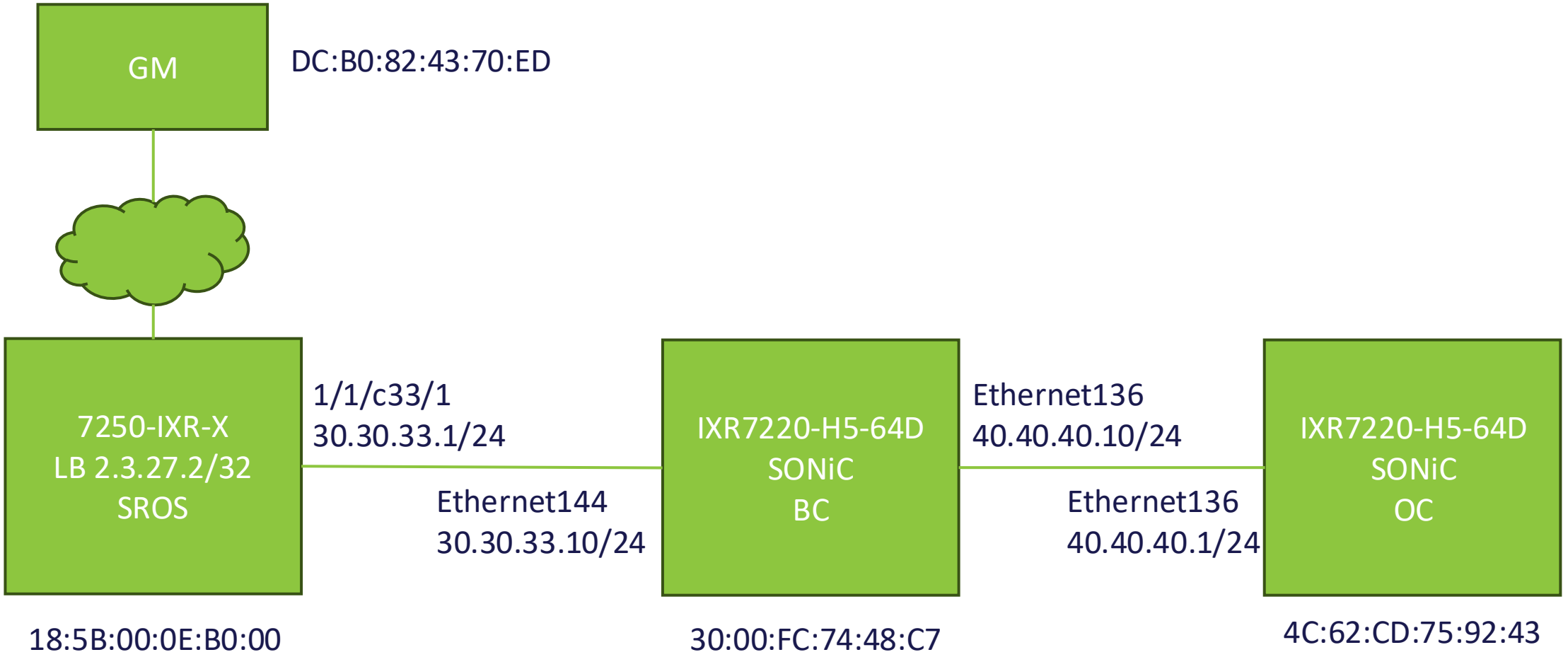
Switch Abstraction Interface (SAI) for BC

- SAI provides a vendor-independent API for controlling forwarding elements
- It separates the control stack from vendor specific forwarding implementation
- SAI API for BC
- sai_create_hostif_trap:
  - This would create the trap for PTP packets as well as load firmware for PTP.
  - The trap type attribute to be used is SAI_HOSTIF_TRAP_TYPE_PTP
- sai_remove_hostif_trap:
  - This would uninstall the trap for PTP packets as well as unload firmware for PTP.
  - The trap type attribute to be used is SAI_HOSTIF_TRAP_TYPE_PTP
- sai_set_port_attribute:
  - This would be used to enable or disable PHY timestamping for 1-step.
  - The attribute to be used is SAI_PORT_ATTR_PTP_MODE

Onboarding new SONiC Switches for PTP

- PTP docker is controlled by feature flag
  - Platform env config - ptp_enabled
- Common Changes already present
  - Yang/CLI commands for configuring and enabling L3 PTP
  - SWSS changes handle PTP config
  - COPP trap for PTP packets
  - PTP container that runs on supported hardware
    - Includes configuration templates for ptp4l config
    - Telemetry support for status and statistics
- Silicon specific changes
  - SAI Implementation
  - Switch ASIC firmware
  - Kernel drivers that implement PHC APIs and upscale timestamps to PTP stack

Test Setup


GM
DC:B0:82:43:70:ED
7250-IXR-X
LB 2.3.27.2/32
SROS
18:5B:00:0E:B0:00
1/1/c33/1
30.30.33.1/24
Ethernet144
30.30.33.10/24
IXR7220-H5-64D
SONiC
BC
30:00:FC:74:48:C7
Ethernet136
40.40.40.10/24
Ethernet136
40.40.40.1/24
IXR7220-H5-64D
SONiC
OC
4C:62:CD:75:92:43

PTP configuration on SONiC

- BC
  - config feature state ptp enabled
  - config ptp domain-number 44
  - config ptp unicast-master-table add 1
  - config ptp unicast-master-table ip add 1 2.3.27.2
  - config ptp interface add Ethernet144 1
  - config ptp interface add Ethernet136 –unicast-listen
- OC
  - config feature state ptp enabled
  - onfig ptp domain-number 44
  - config ptp unicast-master-table add 1
  - config ptp unicast-master-table ip add 1 40.40.40.10
  - config ptp interface add Ethernet136 1

Design considerations

- It is important to have a good crystal oscillator as a clock source
  - Thermal stability
  - Superior holdover performance
  - Low phase noise and jitter

Call to Action

- Problem to solve
  - PTP Boundary Clock support in open community software with merchant silicon for data center use cases
- Provide feedback on HLD/Code PRs
- Consider migration to precision time (PTP) aware hardware
- Contribute the OCPTAP's knowledge base with your contributions
- Work with SAI & SONiC communities to add support for P2P TC & 2-step BC
- Additional Information
  - OCP TAP Project: https://www.opencompute.org/community/time-appliances-project-tap
  - SAI Project : https://github.com/opencomputeproject/SAI
  - SONiC Project: https://github.com/sonic-net/SONiC/wiki/Sonic-Roadmap-Planning

Thank You!

